



Zwavhudi Andy-Morr Musekene

Final-Year BEngTech Computer Engineering Student Aspiring
Software Engineer | Embedded Systems | Java | SQL | C/C++

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🔗 GitHub

PROFILE

Final-year BEngTech Computer Engineering student at Cape Peninsula University of Technology with experience in software development, embedded systems, networking and databases. Skilled in Java, C/C++, SQL and JavaScript, with practical experience building graphics applications, networking systems and hardware-software integration projects. Passionate about solving real-world problems and seeking a graduate software engineering or embedded systems role.

SKILLS

Languages:

Java
SQL
C/C++
JavaScript
Python (Basic)

Concepts:

Object-Oriented Programming
Data Structures
Operating Systems
Computer Networking
Embedded Systems
Database Design

Technologies:

Git & GitHub
MySQL
Arduino
WebGL
Socket Programming
HTML
Wireshark

PROJECTS

Smart Robotic Arm

- Arduino-based robotic arm using servo motors and flex sensors.
- Programmed servo motors and flex sensors using Arduino.
- Developed gesture-based movement control.
- Tested and calibrated sensor responses using Tinkercad simulation.
- Integrated embedded systems and sensor interfacing.
- Designed and tested system in simulation and hardware.

Maintenance Management Database System

- Designed relational database.
- Implemented SQL queries and normalization.
- Designed normalized relational tables and implemented SQL queries for data retrieval and management.

WebGL Snakes and Ladders

- Built interactive game using JavaScript and WebGL.
- Implemented animations and game logic.
- Implemented player movement logic and turn handling.
- Rendered board graphics using WebGL primitives.

UDP Client-Server Application

- Developed network communication using sockets.
- Implemented packet transmission and message exchange.
- Used Python socket programming to establish client-server communication.
- Tested packet transmission using localhost [↗](#).

PROJECT EXPERIENCE

Embedded

- Gesture-controlled robotic arm.
- IoT monitoring system.
- Smart home automation.

Software

- Maintenance Management System with SQL.
- REST API.
- Portfolio website.
- Java desktop app.
- Full-stack web app.

Networking

- Chat application.
- Packet analyzer.
- File transfer system.

EDUCATION

**Bachelor of Engineering Technology in Computer Engineering,
Cape Peninsula University of Technology (CPUT)**

Relevant Coursework:

- Embedded Systems
- Database Systems
- Networking Systems
- Operating Systems
- Software Design
- Computer Graphics
- Digital Systems

AWARDS

**Member of Golden Key honorary society,
Golden Key honorary society**
2025

Provincial colours, Limpopo Hockey

In the year 2023 i was called up to go represent my province in Kwazulu-Natal in the regional tournament

LANGUAGES

English	
Afrikaans	
Tshivenda	
Sepedi	
isiXhosa	